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**First Variation Formula For Area of Submanifolds:
An Exposition**

Mid-Semester Report in Geometric Measure Theory (Fourth Semester)

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Date: March 24, 2026

Abstract

This report provides an exposition of a central result in the calculus of variations and the theory of minimal surfaces, namely the first variation formula for area of submanifolds. This result describes the rate of change of the area of a submanifold under a small perturbation and relates it to a natural geometric quantity, the mean curvature vector. To arrive at this formula, we must first establish a rigorous notion of area for submanifolds. To this end, we first establish via the classical volume formula that the area form on a submanifold is equivalent to the n -dimensional Hausdorff measure. We then study Lipschitz functions on submanifolds, and arrive at an area formula for Lipschitz functions defined on submanifolds. Finally, we shall derive the first variation formula and explain how it leads naturally to the notion of a minimal surface as a surface with vanishing mean curvature.

Contents

0	Introduction	3
1	Geometric Preliminaries on Submanifolds	5
1.1	Submanifolds of $\mathbb{R}^{n+k}$ and Hausdorff Measure	5
1.2	Lipschitz Maps on Submanifolds and the Area Formula	8
1.3	The Second Fundamental Form and Curvature	11
2	First Variation Formula for Area of Submanifolds	14
2.1	Perturbations of Submanifolds and Preliminary Lemmas	14
2.2	Proof of The First Variation Formula	17
2.3	Minimal Surfaces and other Applications of First Variation	21

0 Introduction

The aim of this report is to present an exposition of an important result in the calculus of variations and differential geometry, namely the first variation formula for area of a submanifold. Broadly speaking, we would like to understand how the area of a geometric object changes under a small perturbation of the manifold.

More precisely, the central question we seek to address is the following.

Question 0.1. Let M be an n -dimensional C^2 submanifold of $\mathbb{R}^{n+k}$ and let

$$\Phi : (-\varepsilon, \varepsilon) \times M \rightarrow \mathbb{R}^{n+k}$$

be a perturbation of M such that $\Phi_0(x) = x \ \forall x \in M$, $\Phi_t(x) = x$ for all $x \in M \setminus K$, where $K \subset M$ is compact, $\frac{\partial \Phi}{\partial t}$ is C^1 on $(-\varepsilon, \varepsilon) \times M$, and finally Φ_t is injective. Can we relate the rate of change of area at $t = 0$ to a meaningful geometric quantity? In other words, is there a natural expression for

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\Phi_t(K)),$$

where $\mathcal{H}^n$ denotes the n -dimensional Hausdorff measure in $\mathbb{R}^{n+k}$?

The first variation formula provides an affirmative answer to this question. As we shall see, the rate of change of area can be expressed in terms of intrinsic geometric quantities associated with the submanifold M , most notably its *mean curvature*.

Before studying how this quantity relates to the geometry of M , we must first address a foundational question: why is the Hausdorff measure $\mathcal{H}^n$ a meaningful way to measure the area of subsets of M ? This will be discussed in Section 1. We shall recall the classical area formula, and show that the natural volume form on a submanifold $M \subset \mathbb{R}^{n+k}$ (viewed as a Riemannian manifold with the metric induced from the Euclidean inner product) coincides with the n -dimensional Hausdorff measure $\mathcal{H}^n$; see Theorem 1.6.

After establishing this correspondence, we study Lipschitz maps on submanifolds, and derive an analogous area formula for Lipschitz maps defined on submanifolds. We then introduce several geometric notions associated with submanifolds of $\mathbb{R}^{n+k}$, including the gradient of functions, the divergence of vector fields, and most importantly the *second fundamental form* and the curvature of M . Since our manifolds are embedded in Euclidean space, it will be sufficient for our purposes to define these objects extrinsically, which considerably simplifies the discussion. This approach does not sacrifice generality: by the Nash embedding theorem, every Riemannian manifold can be isometrically embedded into Euclidean space.

With these preliminaries in place, Section 2 focuses on the variational aspects. After proving several preliminary lemmas, we derive the first variation formula for submanifolds.

This gives an answer to our central question as the formula states that (see Theorem 2.8)

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\Phi_t(K)) = \int_M \operatorname{div}_M X \, d\mathcal{H}^n = - \int_M \langle X, H \rangle \, d\mathcal{H}^n,$$

where H denotes the mean curvature vector of M , and X is the velocity vector field associated with the perturbation, defined by

$$X(x) = \frac{\partial \Phi}{\partial t}(x, 0)$$

Finally, we conclude the report by discussing an important consequence of the first variation formula, namely a convenient way of defining and understanding *minimal surfaces*. These are submanifolds which locally minimizes area, something that shall be made more precise in the last section. The main reference for this report is [3]

1 Geometric Preliminaries on Submanifolds

1.1 Submanifolds of $\mathbb{R}^{n+k}$ and Hausdorff Measure

In this section, we recall some basic geometric facts concerning C^k submanifolds of $\mathbb{R}^{n+k}$, together with the area formula for Lipschitz maps on $\mathbb{R}^n$. As an important consequence of the area formula, we shall show that the Riemannian volume measure on a submanifold M coincides with the n -dimensional Hausdorff measure $\mathcal{H}^n$ in $\mathbb{R}^{n+k}$.

Definition 1.1 (C^k submanifolds of $\mathbb{R}^{n+k}$). Let $M \subset \mathbb{R}^{n+k}$ be a subset. We say that M is a C^k n -dimensional submanifold of $\mathbb{R}^{n+k}$ if there exists a C^k manifold $\widetilde{M}$ of dimension n together with a map $f : \widetilde{M} \rightarrow \mathbb{R}^{n+k}$ which is a homeomorphism onto its image and an immersion (i.e. $df_x : T_x \widetilde{M} \rightarrow \mathbb{R}^{n+k}$ is injective for every $x \in \widetilde{M}$), such that $M = f(\widetilde{M})$.

Equivalently, by the implicit function theorem, $M \subset \mathbb{R}^{n+k}$ is a C^k n -dimensional submanifold if and only if for every $x \in M$ there exist an open neighbourhood $U \subset \mathbb{R}^{n+k}$, $x \in U$, and a C^k diffeomorphism $\Phi : U \rightarrow V \subset \mathbb{R}^{n+k}$ such that

$$\Phi(U \cap M) = \{(y_1, \dots, y_{n+k}) \in V : y_{n+1} = \dots = y_{n+k} = 0\}.$$

Furthermore, we can also show via the inverse function theorem, this is equivalent to the condition that for all $x \in M$, $\exists$ a chart $\varphi : y \in W \subset \mathbb{R}^n \rightarrow \varphi(W) \subset M$ s.t. $\varphi(y) = x$, φ is C^k , $d\varphi|_x$ is injective, and φ is a homeomorphism onto its image.

Remark 1.2 (Induced Riemannian structure). For each $x \in M$, the tangent space $T_x M$ naturally identifies with an n -dimensional subspace of $\mathbb{R}^{n+k}$. The Euclidean inner product of $\mathbb{R}^{n+k}$ therefore induces a canonical inner product on $T_x M$ given by

$$\langle v, w \rangle_{T_x M} := \langle v, w \rangle_{\mathbb{R}^{n+k}}, \quad v, w \in T_x M.$$

This endows M with a natural Riemannian metric. Henceforth, whenever we consider a submanifold $M \subset \mathbb{R}^{n+k}$, it will always be understood as a Riemannian manifold equipped with this induced inner product.

Lemma 1.3 (Local orthonormal frames for $T_y M$ and $(T_y M)^\perp$). Let $M \subset \mathbb{R}^{n+k}$ be a C^2 n -dimensional submanifold and let $y_0 \in M$. Then there exists a neighbourhood $U \subset M$ of y_0 and C^1 vector fields

$$\tau_1, \dots, \tau_n, \eta_1, \dots, \eta_k$$

defined on U such that for every $y \in U$:

- (i) $\{\tau_1(y), \dots, \tau_n(y)\}$ is an orthonormal basis of $T_y M$,
- (ii) $\{\eta_1(y), \dots, \eta_k(y)\}$ is an orthonormal basis of $(T_y M)^\perp$,
- (iii) $\{\tau_1(y), \dots, \tau_n(y), \eta_1(y), \dots, \eta_k(y)\}$ is an orthonormal basis of $\mathbb{R}^{n+k}$.

Proof. Since M is C^2 , there exists a local parametrisation $\Psi : V \subset \mathbb{R}^n \rightarrow \mathbb{R}^{n+k}$ which is C^2 such that $\Psi(x_0) = y_0$ and $\Psi(V)$ is a neighbourhood of y_0 in M .

Then the vectors $D_1\Psi(x), \dots, D_n\Psi(x)$ form a basis of $T_{\Psi(x)}M$ and depend C^1 on x .

Choose vectors $\xi_1, \dots, \xi_k \in \mathbb{R}^{n+k}$ such that $D_1\Psi(x_0), \dots, D_n\Psi(x_0), \xi_1, \dots, \xi_k$ are linearly independent in $\mathbb{R}^{n+k}$. By continuity of the determinant, shrinking V if necessary, these vectors remain linearly independent for all $x \in V$.

Applying the Gram–Schmidt orthogonalisation process to the C^1 vector fields

$$D_1\Psi(x), \dots, D_n\Psi(x), \xi_1, \dots, \xi_k,$$

we obtain C^1 orthonormal vector fields $\tau_1, \dots, \tau_n, \eta_1, \dots, \eta_k$ on $\Psi(V)$ satisfying (i)–(iii). $\square$

Recall 1.4 (Area Formula for Lipschitz Maps on $\mathbb{R}^n$). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be Lipschitz with $n \leq m$, and let $A \subset \mathbb{R}^n$ be $\mathcal{L}^n$ -measurable. Then, for any $\mathcal{L}^n$ -integrable function $g : A \rightarrow \mathbb{R}$,

$$\int_A g(x) Jf(x) d\mathcal{L}^n(x) = \int_{\mathbb{R}^m} \sum_{x \in A \cap f^{-1}(y)} g(x) d\mathcal{H}^n(y).$$

If f is injective, we can reformulate the above and get the more familiar form of change of variables for $g : \mathbb{R}^m \rightarrow \mathbb{R}$ integrable.

$$\int_{\mathbb{R}^n} g(f(x)) Jf(x) d\mathcal{L}^n(x) = \int_{\mathbb{R}^m} g(x) d\mathcal{H}^n(y). \quad (1)$$

In particular, taking $g \equiv 1$ on A when f is injective, we obtain

$$\mathcal{H}^n(f(A)) = \int_A Jf(x) d\mathcal{L}^n(x).$$

Recall 1.5 (Riemannian volume). Let (M, g) be an oriented n -dimensional C^k Riemannian manifold. We briefly recall how volume in Riemannian geometry is defined.

Note that there exists a unique smooth n -form dvol_g (the *Riemannian volume form*) characterized by the property that for every $x \in M$ and every positively oriented orthonormal basis $(e_1|_x, \dots, e_n|_x)$ of $T_x M$,

$$\text{dvol}_g|_x(e_1|_x, \dots, e_n|_x) = 1.$$

If $\varphi : U \subset \mathbb{R}^n \rightarrow M$ is a positively oriented chart, then on U we have

$$\varphi^*(\text{dvol}_g) = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n,$$

where $dx^1 \wedge \dots \wedge dx^n$, is the standard n -form on $\mathbb{R}^n$, and

$$g_{ij}(x) = \left\langle D\varphi(x)(e_i|_x), D\varphi(x)(e_j|_x) \right\rangle_{\mathbb{R}^{n+k}}.$$

If $A \subset U$ is measurable, we define

$$\text{Vol}_g(\varphi(A)) = \int_A \sqrt{\det(g_{ij}(x))} dx.$$

Using a partition of unity, this definition extends to all compactly supported measurable subsets of M .

More generally, one obtains a Borel measure Vol_g on M characterized by the property that whenever A is contained in a chart,

$$\text{Vol}_g(A) = \int_A d\text{vol}_g.$$

It is straightforward to verify that this well defined.

We now prove that the n -dimensional Hausdorff measure on M coincides with the Riemannian volume measure.

Theorem 1.6 (Equivalence of Hausdorff measure and Riemannian volume). *Let $M \subset \mathbb{R}^{n+k}$ be an n -dimensional C^1 submanifold, equipped with the Riemannian metric induced by the Euclidean inner product. Let Vol_g denote the associated Riemannian volume measure on M .*

Then for every Borel set $A \subset M$, $\text{Vol}_g(A) = \mathcal{H}^n(A)$.

Proof. Since M is second countable, it admits a countable atlas. Hence we may write any Borel set $A \subset M$ as a disjoint union $A = \bigsqcup_{i=1}^{\infty} A_i$, where each A_i is Borel and contained in the image of a single chart.

Both Vol_g and $\mathcal{H}^n$ are Borel measures on M . Therefore, by countable additivity, it suffices to prove the equality

$$\text{Vol}_g(A_i) = \mathcal{H}^n(A_i)$$

whenever A_i is contained in one chart.

Thus, assume $A \subset \varphi(U)$, where $\varphi : U \subset \mathbb{R}^n \rightarrow M$ is a C^1 parametrization. Then $A = \varphi(B)$ for some Borel set $B \subset U$.

By the area formula,

$$\mathcal{H}^n(A) = \mathcal{H}^n(\varphi(B)) = \int_B J\varphi(x) dx.$$

Note that:

$$J\varphi(x) = \sqrt{\det(D\varphi(x)^T D\varphi(x))} = \sqrt{\det(\langle D\varphi(x)(e_i|_x), D\varphi(x)(e_j|_x) \rangle)}.$$

But the matrix

$$g_{ij}(x) = \langle D\varphi_x(e_i|_x), D\varphi_x(e_j|_x) \rangle$$

is precisely the coordinate representation of the induced Riemannian metric. Therefore,

$$J\varphi(x) = \sqrt{\det(g_{ij}(x))}.$$

By definition of the Riemannian volume measure,

$$\text{Vol}_g(A) = \int_B \sqrt{\det(g_{ij}(x))} dx.$$

Combining the previous identities gives $\mathcal{H}^n(A) = \text{Vol}_g(A)$. □

1.2 Lipschitz Maps on Submanifolds and the Area Formula

In this section we study Lipschitz maps $f : M \rightarrow \mathbb{R}^\ell$, where $M \subset \mathbb{R}^{n+k}$ is an n -dimensional C^1 submanifold. We prove that such maps are differentiable $\mathcal{H}^n$ -almost everywhere on M , in the sense that for $\mathcal{H}^n$ -a.e. $x \in M$ there exists a linear map $df|_x : T_x M \rightarrow T_{f(x)} \mathbb{R}^\ell \simeq \mathbb{R}^\ell$. We also derive an analogue of the area formula for Lipschitz maps defined on submanifolds.

Lemma 1.7. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be Lipschitz and let $0 \leq s < \infty$. Then for every $A \subset \mathbb{R}^n$,*

$$\mathcal{H}^s(f(A)) \leq \text{Lip}(f)^s \mathcal{H}^s(A),$$

where

$$\text{Lip}(f) := \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|} < \infty.$$

In particular, if $\mathcal{H}^s(A) = 0$, then $\mathcal{H}^s(f(A)) = 0$.

Proof. Recall that for $\delta > 0$, the s -dimensional Hausdorff δ -content of A is defined by

$$\mathcal{H}_\delta^s(A) = \inf \left\{ \sum_i \alpha(s) \left(\frac{\text{diam}(C_i)}{2} \right)^s : A \subset \bigcup_i C_i, \text{diam}(C_i) \leq \delta \right\},$$

and

$$\mathcal{H}^s(A) = \lim_{\delta \downarrow 0} \mathcal{H}_\delta^s(A).$$

Let $\{C_i\}$ be a cover of A with $\text{diam}(C_i) \leq \delta$. Then $\{f(C_i)\}$ is a cover of $f(A)$ and, since f is Lipschitz,

$$\text{diam}(f(C_i)) \leq \text{Lip}(f) \text{diam}(C_i).$$

Hence

$$\sum_i \alpha(s) \left(\frac{\text{diam}(f(C_i))}{2} \right)^s \leq \text{Lip}(f)^s \sum_i \alpha(s) \left(\frac{\text{diam}(C_i)}{2} \right)^s.$$

Taking the infimum over all such covers of A , we obtain

$$\mathcal{H}_{\text{Lip}(f)\delta}^s(f(A)) \leq \text{Lip}(f)^s \mathcal{H}_\delta^s(A).$$

Letting $\delta \downarrow 0$ yields $\mathcal{H}^s(f(A)) \leq \text{Lip}(f)^s \mathcal{H}^s(A)$. □

Theorem 1.8 (Almost everywhere differentiability). *Let $M \subset \mathbb{R}^{n+k}$ be an n -dimensional C^1 submanifold and let $f : M \rightarrow \mathbb{R}^\ell$ be Lipschitz. Then for $\mathcal{H}^n$ -a.e. $x \in M$ $df|_x : T_x M \rightarrow \mathbb{R}^\ell$ exists.*

Proof. Let $\varphi : U \subset \mathbb{R}^n \rightarrow M$ be a C^1 chart. By shrinking U if necessary, we may assume U is bounded. Since φ is C^1 on a bounded set, it is Lipschitz. Hence the composition

$$F := f \circ \varphi : U \rightarrow \mathbb{R}^\ell$$

is Lipschitz.

Let $p \in U$ be a point where $F = f \circ \varphi$ is differentiable, and set $x = \varphi(p)$.

Recall that

$$T_x M = \text{span}\{D\varphi_p(e_1|_p), \dots, D\varphi_p(e_n|_p)\}.$$

If $v \in T_x M$, then there exists $c \in \mathbb{R}^n$ such that $v = D\varphi_p(c)$.

Recall that, if it exists,

$$df|_x(v) := \left. \frac{d}{dt} \right|_{t=0} f(\alpha(t)),$$

whenever α is a C^1 curve in M satisfying $\alpha(0) = x$ and $\alpha'(0) = v$.

To compute this derivative, consider the curve $\alpha(t) := \varphi(p + tc)$. Then $\alpha(0) = x$ and $\alpha'(0) = D\varphi_p(c) = v$.

Since F is differentiable at p ,

$$\left. \frac{d}{dt} \right|_{t=0} f(\alpha(t)) = \left. \frac{d}{dt} \right|_{t=0} F(p + tc) = DF(p)c.$$

Thus the derivative exists and $df|_x(v) = DF(p)c$.

Let $A := \{p \in U : F \text{ is not differentiable at } p\}$. By Rademacher's theorem, since F is Lipschitz, F is differentiable $\mathcal{L}^n$ -a.e. on U and thus $\mathcal{L}^n(A) = 0$.

From the previous argument, $df|_x$ exists for all $x \in \varphi(U) \setminus \varphi(A)$. Since φ is Lipschitz on U , Lemma 1.7 implies $\mathcal{H}^n(\varphi(A)) = 0$. Therefore $df|_x$ exists for $\mathcal{H}^n$ -a.e. $x \in \varphi(U)$.

Finally, since M admits a countable atlas $M = \bigcup_{i=1}^\infty \varphi_i(U_i)$, and the union of countably many $\mathcal{H}^n$ -null sets is again $\mathcal{H}^n$ -null, it follows that $df|_x$ exists for $\mathcal{H}^n$ -a.e. $x \in M$. □

Remark 1.9. We shall use the notation $D_v f(x)$ and $df|_x(v)$ interchangeably, as sometimes the former may be more convenient.

Lemma 1.10. *Let $M \subset \mathbb{R}^{n+k}$ be an n -dimensional C^1 submanifold and let $f : M \rightarrow \mathbb{R}^\ell$ be Lipschitz. Let $\varphi : U \subset \mathbb{R}^n \rightarrow M$ be a C^1 chart.*

Define the Jacobian of f by

$$J^M f(x) := \sqrt{\det((df|_x)^T df|_x)},$$

whenever $df|_x$ exists.

Then for $\mathcal{L}^n$ -a.e. $p \in U$, $J(f \circ \varphi)(p) = J^M f(\varphi(p)) J\varphi(p)$.

Proof. At points where p where $df|_{\varphi(p)}$ exists, $d(f \circ \varphi)_p = df|_{\varphi(p)} \circ d\varphi|_p$.

Hence

$$(d(f \circ \varphi)|_p)^T d(f \circ \varphi)|_p = d\varphi|_p^T (df|_{\varphi(p)})^T df|_{\varphi(p)} d\varphi|_p.$$

Taking determinants,

$$\det\left((d(f \circ \varphi)|_p)^T d(f \circ \varphi)|_p\right) = \det(d\varphi|_p^T) \det\left((df|_{\varphi(p)})^T df|_{\varphi(p)}\right) \det(d\varphi|_p).$$

Taking square roots yields

$$J(f \circ \varphi)(p) = J^M f(\varphi(p)) J\varphi(p).$$

□

Theorem 1.11 (Area formula on submanifolds). *Let $M \subset \mathbb{R}^{n+k}$ be an n -dimensional C^1 submanifold and let $f : M \rightarrow \mathbb{R}^m$ be Lipschitz with $m \geq n$. Let $A \subset M$ be Borel.*

Then

$$\int_A J^M f(x) d\mathcal{H}^n(x) = \int_{\mathbb{R}^m} \mathcal{H}^0(A \cap f^{-1}(y)) d\mathcal{H}^n(y).$$

Proof. First assume that $A \subset \varphi(U)$, where $\varphi : U \subset \mathbb{R}^n \rightarrow M$ is a C^1 chart, where U is bounded

Since φ is Lipschitz, the classical area formula holds. Thus, taking $g(x) = J^M f(x)$ on A , and 0 on $\mathbb{R}^{n+k} \setminus A$, and $f = \varphi$ in Equation (1), we get:

$$\int_A J^M f(x) d\mathcal{H}^n(x) = \int_{\varphi^{-1}(A)} J^M f(\varphi(p)) J\varphi(p) dp.$$

By the previous lemma,

$$J^M f(\varphi(p)) J\varphi(p) = J(f \circ \varphi)(p).$$

Hence

$$\int_A J^M f(x) d\mathcal{H}^n(x) = \int_{\varphi^{-1}(A)} J(f \circ \varphi)(p) dp.$$

Since $f \circ \varphi$ is Lipschitz, applying the classical area formula again yields

$$\int_{\varphi^{-1}(A)} J(f \circ \varphi)(p) dp = \int_{\mathbb{R}^m} \mathcal{H}^0(\varphi^{-1}(A) \cap (f \circ \varphi)^{-1}(y)) d\mathcal{H}^n(y).$$

Because φ is injective,

$$\mathcal{H}^0(\varphi^{-1}(A) \cap (f \circ \varphi)^{-1}(y)) = \mathcal{H}^0(A \cap f^{-1}(y)).$$

This proves the formula when A is contained in one chart.

For a general Borel set $A \subset M$, write A as a disjoint union of countably many sets each contained in a chart and use countable additivity on both sides. $\square$

1.3 The Second Fundamental Form and Curvature

In this section, we introduce several definitions that arise naturally on submanifolds, such as the gradient and divergence. We also study the curvature of submanifolds through the second fundamental form.

Definition 1.12 (Gradient and divergence on a submanifold). Let $M \subset \mathbb{R}^{n+k}$ be a C^1 submanifold and let $f : M \rightarrow \mathbb{R}$ be a function such that $df|_x$ exists.

The *gradient of f on M* at $x \in M$, denoted $\nabla^M f(x)$, is the unique vector in $T_x M$ such that

$$\langle \nabla^M f(x), v \rangle = df|_x(v) \quad \text{for all } v \in T_x M.$$

If $\{v_1, \dots, v_n\}$ is an orthonormal basis of $T_x M$, then

$$\nabla^M f(x) = \sum_{i=1}^n df|_x(v_i) v_i.$$

A *vector field* on M will simply mean a map $X : M \rightarrow \mathbb{R}^{n+k}$. Note that we do not require $X(x) \in T_x M$.

Write $X = (X_1, \dots, X_{n+k})$ with $X_i : M \rightarrow \mathbb{R}$. The *divergence of X on M* is defined by

$$\operatorname{div}_M X := \sum_{i=1}^{n+k} \langle \nabla^M X_i, e_i \rangle,$$

where $\{e_1, \dots, e_{n+k}\}$ is the standard basis of $\mathbb{R}^{n+k}$.

Equivalently, if $\{v_1, \dots, v_n\}$ is an orthonormal basis of $T_x M$, then

$$\begin{aligned} \operatorname{div}_M X(x) &= \sum_{i=1}^{n+k} \langle \nabla^M X_i(x), e_i \rangle \\ &= \sum_{i=1}^{n+k} \left\langle \sum_{j=1}^n dX_i|_x(v_j) v_j, e_i \right\rangle \\ &= \sum_{j=1}^n \sum_{i=1}^{n+k} dX_i|_x(v_j) \langle v_j, e_i \rangle \\ &= \sum_{j=1}^n \langle (dX_1|_x(v_j), \dots, dX_{n+k}|_x(v_j)), v_j \rangle \\ &= \sum_{j=1}^n \langle dX|_x(v_j), v_j \rangle. \end{aligned}$$

Lemma 1.13 (Curvature decomposition). *Let $M \subset \mathbb{R}^{n+k}$ be a C^2 submanifold and let $\alpha : I \rightarrow M$ be a C^2 curve, where $I \subset \mathbb{R}$ is an interval, such that*

$$\alpha(0) = x, \quad \dot{\alpha}(0) = v \in T_x M.$$

Assume that α is parametrised by arc length, i.e. $|\dot{\alpha}(t)| = 1$.

Let $\{\eta_1, \dots, \eta_k\}$ be a local orthonormal frame for $(T_y M)^\perp$ defined in a neighbourhood of x . Then

$$\alpha''(0) = - \sum_{i=1}^k \langle d\eta_i|_x(v), v \rangle \eta_i(x).$$

Proof. Since $\alpha(t) \in M$ for all t and η_i is normal to M , we have

$$\langle \eta_i(\alpha(t)), \dot{\alpha}(t) \rangle = 0 \quad \text{for all } t.$$

Differentiating with respect to t gives

$$\frac{d}{dt} \langle \eta_i(\alpha(t)), \dot{\alpha}(t) \rangle = \langle d\eta_i|_{\alpha(t)}(\dot{\alpha}(t)), \dot{\alpha}(t) \rangle + \langle \eta_i(\alpha(t)), \alpha''(t) \rangle = 0.$$

Evaluating at $t = 0$ yields

$$\langle \eta_i(x), \alpha''(0) \rangle = - \langle d\eta_i|_x(v), v \rangle.$$

Since $|\dot{\alpha}(t)| = 1$, we have

$$\langle \alpha''(t), \dot{\alpha}(t) \rangle = 0,$$

so $\alpha''(0)$ is orthogonal to $T_x M$. Hence $\alpha''(0) \in (T_x M)^\perp$.

Writing $\alpha''(0)$ in the orthonormal basis $\{\eta_1(x), \dots, \eta_k(x)\}$ of $(T_x M)^\perp$,

$$\alpha''(0) = - \sum_{i=1}^k \langle \alpha''(0), \eta_i(x) \rangle \eta_i(x),$$

□

The previous lemma suggests a natural way to measure the curvature of the submanifold. Indeed, if α is a curve in M with $\alpha(0) = x$ and $\alpha'(0) = v$, then the vector $\alpha''(0)$ can be thought of as measuring the curvature of M in the direction v .

Motivated by this observation, we introduce the following definition.

Definition 1.14 (Second fundamental form). Let $M \subset \mathbb{R}^{n+k}$ be a C^2 submanifold and let $x \in M$. The *second fundamental form* of M at x is the bilinear map

$$B_x : T_x M \times T_x M \longrightarrow (T_x M)^\perp$$

defined by

$$B_x(v_1, v_2) = - \sum_{i=1}^k \langle D_{v_1} \eta_i(x), v_2 \rangle \eta_i(x),$$

where $v_1, v_2 \in T_x M$, $\{\eta_1, \dots, \eta_k\}$ is a local orthonormal frame of $(T_y M)^\perp$ defined in a neighbourhood of x .

The definition is independent of the choice of orthonormal frame of the normal bundle, since by the previous lemma $B_x(v, v) = \alpha''(0)$, which is independent of the choice of frame.

Definition 1.15 (Mean curvature). Let $M \subset \mathbb{R}^{n+k}$ be a C^2 submanifold and let $\{v_1, \dots, v_n\}$ be an orthonormal basis of $T_x M$. The *mean curvature vector* at x is defined by

$$H(x) := \sum_{i=1}^n B_x(v_i, v_i).$$

Using the definition of the second fundamental form, we obtain

$$\begin{aligned} H(x) &= \sum_{i=1}^n B_x(v_i, v_i) \\ &= - \sum_{i=1}^n \sum_{j=1}^k \langle d\eta_j|_x(v_i), v_i \rangle \eta_j \\ &= - \sum_{j=1}^k \left(\sum_{i=1}^n \langle d\eta_j|_x(v_i), v_i \rangle \right) \eta_j \\ &= - \sum_{j=1}^k \operatorname{div}_M(\eta_j) \eta_j. \end{aligned}$$

2 First Variation Formula for Area of Submanifolds

2.1 Perturbations of Submanifolds and Preliminary Lemmas

In this section we introduce the notion of perturbations (or variations) of a submanifold and establish several preliminary lemmas that will be useful in the derivation of the first variation formula for the area of a submanifold.

Definition 2.1 (Perturbations of a submanifold). Let $M \subset \mathbb{R}^{n+k}$ be a C^2 n -dimensional submanifold.

A map $\Phi : (-\varepsilon, \varepsilon) \times M \longrightarrow \mathbb{R}^{n+k}$ is called a *perturbation* (or *variation*) of M if:

- (i) $\frac{\partial \Phi}{\partial t} : (-\varepsilon, \varepsilon) \times M \rightarrow \mathbb{R}^{n+k}$ is of class C^1 ,
- (ii) $\Phi(0, x) = x$ for all $x \in M$ and $\Phi(t, \cdot)$ is injective for each t .

The perturbation is said to be *compactly supported* (or a *compact perturbation*) if there exists a compact set $K \subset M$ such that

$$\Phi(t, x) = x \quad \text{for all } x \in M \setminus K \text{ and all } t \in (-\varepsilon, \varepsilon).$$

Lemma 2.2. Let $M \subset \mathbb{R}^{n+k}$ be a C^1 n -dimensional submanifold, and let $f : (-\varepsilon, \varepsilon) \times M \longrightarrow M$ be a C^1 map. For each $t \in (-\varepsilon, \varepsilon)$, define $f^t(x) := f(t, x)$.

Then the map

$$(t, x) \longmapsto \|Df^t|_x\|$$

is continuous on $(-\varepsilon, \varepsilon) \times M$, where $D_x f^t(x) : T_x M \rightarrow T_{f^t(x)} M$ denotes the derivative with respect to the manifold variable x , and $\|\cdot\|$ denotes the operator norm.

Proof. The claim is local, so let $x_0 \in M$. Choose charts

$$\varphi : U \subset \mathbb{R}^n \rightarrow M, \quad \psi : V \subset \mathbb{R}^n \rightarrow M$$

such that $x_0 = \varphi(p_0)$ and $f(t_0, x_0) \in \psi(V)$ for some $(t_0, x_0) \in (-\varepsilon, \varepsilon) \times M$.

Define the local representation

$$F(t, p) := \psi^{-1}(f(t, \varphi(p))), \quad (t, p) \in (-\varepsilon, \varepsilon) \times U.$$

Since f , φ , and ψ^{-1} are C^1 , it follows that F is a C^1 map.

For $x = \varphi(p)$, the chain rule gives

$$D_x f^t(\varphi(p)) = D\psi_{F(t,p)} \circ D_p F(t, p) \circ D\varphi_p^{-1}.$$

Taking operator norms, we obtain

$$\|D_x f^t(\varphi(p))\| = \|D\psi_{F(t,p)} \circ D_p F(t, p) \circ D\varphi_p^{-1}\|.$$

Since F is C^1 and the derivatives of the charts are continuous, the map

$$(t, p) \mapsto \|D_x f^t(\varphi(p))\|$$

is continuous on $(-\varepsilon, \varepsilon) \times U$.

Since φ is any arbitrary chart, it follows that:

$$(t, x) \mapsto \|D_x f^t(x)\|$$

is continuous on $(-\varepsilon, \varepsilon) \times M$. □

Lemma 2.3. *Let $M \subset \mathbb{R}^{n+k}$ be a C^2 submanifold and let $\Phi : (-\varepsilon, \varepsilon) \times M \rightarrow \mathbb{R}^{n+k}$ be a compact perturbation of M , i.e. $\exists K$ compact s.t. $\Phi(x, t) = x$ for all $x \in M \setminus K$. Then Φ is of class C^1 on $(-\varepsilon, \varepsilon) \times M$, and writing $X(x) := \left. \frac{\partial \Phi}{\partial t}(t, x) \right|_{t=0}$, we may express Φ in the form*

$$\Phi(t, x) = x + t(X(x) + E(t, x)),$$

where $E : (-\varepsilon, \varepsilon) \times M \rightarrow \mathbb{R}^{n+k}$ is continuous and satisfies

$$\lim_{t \rightarrow 0} \sup_{x \in K} \left(|E(t, x)| + \sup_{\substack{\tau \in T_x M \\ |\tau| \leq 1}} |D_\tau E(t, x)| \right) = 0.$$

Equivalently,

$$\sup_{x \in K} \left(|E(t, x)| + \|D_x E(t, x)\| \right) \rightarrow 0 \quad \text{as } t \rightarrow 0.$$

Proof. For $(t, x) \in (-\varepsilon, \varepsilon) \times M$, we write

$$\Phi(t, x) = x + \int_0^t \frac{\partial \Phi}{\partial s}(s, x) ds.$$

and by definition of X , we obtain

$$\Phi(t, x) = x + tX(x) + \int_0^t \left(\frac{\partial \Phi}{\partial s}(s, x) - \frac{\partial \Phi}{\partial s}(0, x) \right) ds.$$

Define

$$E(t, x) := \frac{1}{t} \int_0^t \left(\frac{\partial \Phi}{\partial s}(s, x) - \frac{\partial \Phi}{\partial s}(0, x) \right) ds \quad (t \neq 0),$$

and set $E(0, x) = 0$. Then

$$\Phi(t, x) = x + t(X(x) + E(t, x)).$$

Since $\frac{\partial \Phi}{\partial t}$ is C^1 , the map

$$(s, x) \mapsto \frac{\partial \Phi}{\partial s}(s, x)$$

is continuous. Hence it is uniformly continuous on $[-\varepsilon_0, \varepsilon_0] \times K$ for any $0 < \varepsilon_0 < \varepsilon$.

Given $\alpha > 0$, we can therefore choose $\delta > 0$ such that

$$\left| \frac{\partial \Phi}{\partial s}(s, x) - \frac{\partial \Phi}{\partial s}(0, x) \right| < \alpha$$

whenever $|s| < \delta$ and $x \in K$. Thus for $|t| < \delta$,

$$|E(t, x)| \leq \frac{1}{|t|} \int_0^{|t|} \alpha \, ds = \alpha,$$

and hence

$$\sup_{x \in K} |E(t, x)| \rightarrow 0 \quad \text{as } t \rightarrow 0.$$

Next we estimate the derivative in the manifold variable. Let $v \in T_x M$ with $|v| = 1$. Since $\frac{\partial \Phi}{\partial s}$ is of class C^1 , we may differentiate under the integral sign by DCT to obtain

$$D_x E(t, x)(v) = \frac{1}{t} \int_0^t \left(D_x \left(\frac{\partial \Phi}{\partial s}(s, x) \right) (v) - D_x \left(\frac{\partial \Phi}{\partial s}(0, x) \right) (v) \right) ds.$$

Taking absolute values and using the definition of the operator norm gives

$$|D_x E(t, x)(v)| \leq \frac{1}{|t|} \int_0^{|t|} \left\| D_x \left(\frac{\partial \Phi}{\partial s}(s, x) \right) - D_x \left(\frac{\partial \Phi}{\partial s}(0, x) \right) \right\| ds.$$

Taking the supremum over all $v \in T_x M$ with $|v| = 1$ yields

$$\|D_x E(t, x)\| \leq \frac{1}{|t|} \int_0^{|t|} \left\| D_x \left(\frac{\partial \Phi}{\partial s}(s, x) \right) - D_x \left(\frac{\partial \Phi}{\partial s}(0, x) \right) \right\| ds.$$

By the previous lemma, the map

$$(s, x) \mapsto \left\| D_x \left(\frac{\partial \Phi}{\partial s}(s, x) \right) \right\|$$

is continuous, hence uniformly continuous on $[-\varepsilon_0, \varepsilon_0] \times K$. Therefore the integrand converges uniformly to 0 as $s \rightarrow 0$, and we conclude

$$\sup_{x \in K} \|D_x E(t, x)\| \rightarrow 0 \quad \text{as } t \rightarrow 0.$$

Combining the two estimates gives

$$\sup_{x \in K} \left(|E(t, x)| + \|D_x E(t, x)\| \right) \rightarrow 0 \quad \text{as } t \rightarrow 0,$$

which completes the proof. □

2.2 Proof of The First Variation Formula

Having established all the required prerequisites, in this section, we may now work towards a proof of the first variation formula.

Theorem 2.4. *Let $M \subset \mathbb{R}^{n+k}$ be a C^2 submanifold and let $\Phi : (-\varepsilon, \varepsilon) \times M \rightarrow \mathbb{R}^{n+k}$ be a compactly supported perturbation with associated compact set $K \subset M$, so that $\Phi(t, x) = x$ for all $x \in M \setminus K$. Define*

$$X(x) := \left. \frac{\partial \Phi}{\partial t}(t, x) \right|_{t=0}.$$

Then, for every $x \in M$,

$$\left. \frac{d}{dt} \right|_{t=0} J_\Phi(t, x) = \operatorname{div}_M X(x),$$

where $J_\Phi(t, x)$ is the Jacobian of $\Phi_t := \Phi(t, \cdot)$ with respect to the submanifold M .

Proof. Let $\{v_1, \dots, v_n\}$ be an orthonormal basis of $T_x M$. By definition,

$$\begin{aligned} J_\Phi(t, x) &= \sqrt{\det((D\Phi_t(x))^T D\Phi_t(x))} \\ &= \sqrt{\det(\langle D_{v_i} \Phi_t(x), D_{v_j} \Phi_t(x) \rangle)}. \end{aligned}$$

Set

$$A_x(t)_{ij} := \langle D_{v_i} \Phi_t(x), D_{v_j} \Phi_t(x) \rangle$$

By Lemma 2.3, we can write

$$\Phi_t(x) = x + t(X(x) + E(t, x)),$$

with $E(t, x)$ satisfying $\sup_{x \in K} (|E(t, x)| + \|D_x E(t, x)\|) \rightarrow 0$ as $t \rightarrow 0$.

Thus, for each i ,

$$D_{v_i} \Phi_t(x) = v_i + tD_{v_i} X(x) + tD_{v_i} E(t, x).$$

Hence

$$\begin{aligned} \langle D_{v_i} \Phi_t(x), D_{v_j} \Phi_t(x) \rangle &= \langle v_i + tD_{v_i} X(x) + tD_{v_i} E(t, x), v_j + tD_{v_j} X(x) + tD_{v_j} E(t, x) \rangle \\ &= \langle v_i, v_j \rangle + t\langle D_{v_i} X(x), v_j \rangle + t\langle v_i, D_{v_j} X(x) \rangle \\ &\quad + t\langle D_{v_i} E(t, x), v_j \rangle + t\langle v_i, D_{v_j} E(t, x) \rangle + O(t^2). \end{aligned}$$

Since $\{v_i\}_{i=1}^n$ is an orthonormal basis and $D_v E(t, x) \rightarrow 0$ as $t \rightarrow 0$ by Lemma 2.3, the last three terms are $o(t)$. Therefore

$$\langle D_{v_i} \Phi_t(x), D_{v_j} \Phi_t(x) \rangle = \delta_{ij} + t(\langle D_{v_i} X(x), v_j \rangle + \langle v_i, D_{v_j} X(x) \rangle) + o(t).$$

Define the matrix B_x by

$$(B_x)_{ij} := \langle D_{v_i} X(x), v_j \rangle + \langle v_i, D_{v_j} X(x) \rangle.$$

Then

$$A_x(t) = I + tB_x + o(t)$$

Recall that the partial derivatives of the determinant map $\det : M_n(\mathbb{R}) \rightarrow \mathbb{R}$ satisfy

$$\frac{\partial \det}{\partial a_{ij}}(A) = \text{adj}(A)_{ji},$$

where $\text{adj}(A)$ denotes the adjugate matrix.

Thus, by the chain rule we obtain

$$\left. \frac{d}{dt} \right|_{t=0} \det(A_x(t)) = \sum_{i,j=1}^n \frac{\partial \det}{\partial a_{ij}}(A_x(0)) \left. \frac{d}{dt} \right|_{t=0} (A_x(t))_{ij}.$$

Since $A_x(0) = I$ and $\left. \frac{d}{dt} \right|_{t=0} (A_x(t))_{ij} = (B_x)_{ij}$, this gives

$$\left. \frac{d}{dt} \right|_{t=0} \det(A_x(t)) = \sum_{i,j=1}^n \delta_{ji} B_{x,ij} = \sum_{i=1}^n B_{x,ii} = \text{tr}(B_x).$$

Recalling the definition of divergence, we obtain

$$\text{tr}(B_x) = 2 \sum_{i=1}^n \langle D_{v_i} X(x), v_i \rangle = 2 \text{div}_M X(x).$$

Finally, since $J_\Phi(t, x) = \sqrt{\det(A_x(t))}$, the chain rule gives

$$\left. \frac{d}{dt} \right|_{t=0} J_\Phi(t, x) = \frac{1}{2} \det(A_x(0))^{-1/2} \left. \frac{d}{dt} \right|_{t=0} \det(A_x(t)) = \text{div}_M X(x),$$

as claimed. □

Remark 2.5. In the proof above one obtains in fact a stronger statement: the difference quotient

$$\frac{J_\Phi(t, x) - 1}{t}$$

converges to $\text{div}_M X(x)$ uniformly for $x \in K$ as $t \rightarrow 0$.

Indeed, writing

$$\Phi(t, x) = x + t(X(x) + E(t, x)),$$

Lemma 2.3 shows that

$$\sup_{x \in K} (|E(t, x)| + \|D_x E(t, x)\|) \rightarrow 0 \quad \text{as } t \rightarrow 0.$$

Consequently the expansion

$$\langle D_{v_i} \Phi_t(x), D_{v_j} \Phi_t(x) \rangle = \delta_{ij} + t(\langle D_{v_i} X(x), v_j \rangle + \langle v_i, D_{v_j} X(x) \rangle) + o(t)$$

holds uniformly for $x \in K$. It follows that

$$\frac{J_\Phi(t, x) - 1}{t} \longrightarrow \operatorname{div}_M X(x) \quad \text{uniformly for } x \in K.$$

This uniform convergence will be important in the following theorem as it will justify passing the derivative inside the integral in the area formula when computing the first variation of $\mathcal{H}^n(\Phi_t(M \cap K))$, as will be explained below.

Theorem 2.6. *Let $M \subset \mathbb{R}^{n+k}$ be a C^2 submanifold and let $\Phi : (-\varepsilon, \varepsilon) \times M \rightarrow \mathbb{R}^{n+k}$ be a compact perturbation, i.e. there exists a compact set $K \subset M$ such that*

$$\Phi(t, x) = x \quad \text{for all } x \in M \setminus K \text{ and } t \in (-\varepsilon, \varepsilon).$$

Let

$$X(x) := \left. \frac{\partial \Phi}{\partial t}(t, x) \right|_{t=0}.$$

Then

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\Phi_t(K)) = \int_K \operatorname{div}_M X(x) d\mathcal{H}^n(x).$$

Proof. Since Φ_t is injective and Lipschitz on K , the area formula gives

$$\mathcal{H}^n(\Phi_t(K)) = \int_K J_{\Phi_t}(x) d\mathcal{H}^n(x).$$

Hence

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\Phi_t(K)) = \left. \frac{d}{dt} \right|_{t=0} \int_K J_{\Phi_t}(x) d\mathcal{H}^n(x).$$

By the previous theorem we have

$$\left. \frac{d}{dt} \right|_{t=0} J_{\Phi_t}(x) = \operatorname{div}_M X(x).$$

Moreover, by the remark following that theorem, the difference quotient

$$\frac{J_{\Phi_t}(x) - 1}{t}$$

converges uniformly to $\operatorname{div}_M X(x)$ on K as $t \rightarrow 0$. Therefore we may pass the derivative inside the integral and obtain

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\Phi_t(K)) = \int_K \left. \frac{d}{dt} \right|_{t=0} J_{\Phi_t}(x) d\mathcal{H}^n(x).$$

Substituting the expression for the derivative of the Jacobian gives

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\Phi_t(K)) = \int_K \operatorname{div}_M X(x) d\mathcal{H}^n(x).$$

□

Lemma 2.7. *Let $M \subset \mathbb{R}^{n+k}$ be a C^2 n -dimensional submanifold and let $X : M \rightarrow \mathbb{R}^{n+k}$ be a compactly supported C^1 vector field. Then*

$$\int_M \operatorname{div}_M X d\mathcal{H}^n = - \int_M \langle X, H \rangle d\mathcal{H}^n,$$

where H denotes the mean curvature vector of M .

Proof. Decompose X into its tangential and normal components,

$$X = X_1 + X_2,$$

where $X_1(x) \in T_x M$ and $X_2(x) \in (T_x M)^\perp$.

Since X_1 is tangential and compactly supported, the divergence theorem on M implies

$$\int_M \operatorname{div}_M X_1 d\mathcal{H}^n = 0.$$

Hence

$$\int_M \operatorname{div}_M X = \int_M \operatorname{div}_M X_2.$$

Let $\{\eta_1, \dots, \eta_k\}$ be a local orthonormal frame of the normal bundle near x . Then we may write

$$X_2(x) = \sum_{j=1}^k c_j(x) \eta_j(x)$$

for some scalar functions c_j .

Let $\{v_1, \dots, v_n\}$ be an orthonormal basis of $T_x M$. Then

$$\operatorname{div}_M X_2(x) = \sum_{i=1}^n \langle D_{v_i} X_2(x), v_i \rangle.$$

Differentiating the above expression for X_2 , we obtain

$$D_{v_i} X_2 = \sum_{j=1}^k (D_{v_i} c_j \eta_j + c_j D_{v_i} \eta_j).$$

Taking the inner product with v_i and using that $\langle \eta_j, v_i \rangle = 0$, we obtain

$$\langle D_{v_i} X_2, v_i \rangle = \sum_{j=1}^k c_j \langle D_{v_i} \eta_j, v_i \rangle.$$

Summing over i gives

$$\operatorname{div}_M X_2(x) = \sum_{j=1}^k c_j(x) \left(\sum_{i=1}^n \langle D_{v_i} \eta_j(x), v_i \rangle \right).$$

Recalling the formula for the mean curvature vector

$$H(x) = - \sum_{j=1}^k \left(\sum_{i=1}^n \langle D_{v_i} \eta_j(x), v_i \rangle \right) \eta_j(x),$$

we obtain

$$\operatorname{div}_M X_2(x) = - \langle X_2(x), H(x) \rangle.$$

Integrating over M yields

$$\int_M \operatorname{div}_M X = \int_M \operatorname{div}_M X_2 = - \int_M \langle X_2, H \rangle.$$

Since $H(x) \in (T_x M)^\perp$ and $X_1(x) \in T_x M$, we have $\langle X_1, H \rangle = 0$. Therefore

$$\int_M \langle X_2, H \rangle = \int_M \langle X, H \rangle.$$

Thus

$$\int_M \operatorname{div}_M X \, d\mathcal{H}^n = - \int_M \langle X, H \rangle \, d\mathcal{H}^n.$$

□

Combining our the discussion in this section, we finally obtain the first variation formula of area for submanifolds.

Theorem 2.8 (First variation Formula). *Let $M \subset \mathbb{R}^{n+k}$ be an n -dimensional C^2 submanifold. Let $\{\Phi_t\}_{|t|<\varepsilon}$ be a compactly supported perturbation of M with associated variation vector field*

$$X(x) = \left. \frac{d}{dt} \right|_{t=0} \Phi_t(x).$$

Let $K \subset M$ be a compact set containing the support of X . Then

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\Phi_t(K)) = \int_M \operatorname{div}_M X \, d\mathcal{H}^n = - \int_M \langle X, H \rangle \, d\mathcal{H}^n,$$

where H denotes the mean curvature vector of M .

2.3 Minimal Surfaces and other Applications of First Variation

The first variation formula is a powerful tool with many important geometric consequences. In this section we illustrate some of its basic applications. We begin with a

simple consequence of the first variation, namely that it provides a natural interpretation of the mean curvature flow. We then state a few basic lemmas and use them to see that the first variation leads naturally to the notion of minimal surfaces. Roughly speaking, these are surfaces which locally minimize area; this idea will be made precise in the discussion that follows.

Remark 2.9 (Mean curvature flow and steepest descent of area). Let $\{M_t\}_{t \in \mathbb{R}}$ be a smooth one-parameter family of hypersurfaces in $\mathbb{R}^n$ with parametrization $x(\cdot, t)$. We say that M_t evolves by *mean curvature flow* if

$$\frac{\partial x}{\partial t}(p, t) = \vec{H}(x(p, t)), \quad p \in M_t.$$

We show that this evolution corresponds to moving in the direction in which the area decreases most rapidly.

Recall from the first variation formula that if a hypersurface evolves with velocity vector field X , then

$$\frac{d}{dt} \mathcal{H}^{n-1}(M_t) = - \int_{M_t} \langle X, \vec{H} \rangle d\mathcal{H}^{n-1}.$$

Write $X = X^\perp + X^T$ where $X^\perp$ and X^T denote the normal and tangential components of the velocity. Since $\vec{H}$ is normal to M_t , the tangential component does not contribute, and we obtain

$$\frac{d}{dt} \mathcal{H}^{n-1}(M_t) = - \int_{M_t} \langle X^\perp, \vec{H} \rangle d\mathcal{H}^{n-1}.$$

Among all possible normal velocities $X^\perp$, the quantity on the right is minimized when

$$X^\perp = \vec{H}.$$

Indeed, by the Cauchy–Schwarz inequality,

$$-\langle X^\perp, \vec{H} \rangle \geq -|\vec{H}| |X^\perp|,$$

with equality precisely when $X^\perp$ is parallel to $\vec{H}$.

Thus the choice

$$\frac{\partial x}{\partial t} = \vec{H}$$

gives the direction in which the area functional decreases most rapidly. In this sense, mean curvature flow can be interpreted as the gradient flow of the area functional.

Lemma 2.10. *Let $M \subset \mathbb{R}^{n+k}$ be an n -dimensional C^2 submanifold. Let V_1, V_2 be C^1 tangent vector fields on M . Then the map*

$$x \mapsto B_x(V_1(x), V_2(x))$$

is continuous on M , where B denotes the second fundamental form of M .

Proof. Since M is a C^2 submanifold, around every point of M we can choose a C^1 orthonormal frame of the normal bundle $\{\eta_1, \dots, \eta_k\}$. The second fundamental form can be written as

$$B(V_1, V_2) = - \sum_{i=1}^k \langle V_2, D_{V_1} \eta_i \rangle \eta_i.$$

Because the vector fields η_i are C^1 and V_1 is C^1 , the maps $x \mapsto D_{V_1} \eta_i(x)$ are continuous. Since V_2 is C^1 , it follows that $\langle V_2, D_{V_1} \eta_i \rangle \eta_i$ is continuous and

$$x \mapsto B_x(V_1(x), V_2(x))$$

is continuous. □

Corollary 2.11. *Let $M \subset \mathbb{R}^{n+k}$ be an n -dimensional C^2 submanifold. Then the mean curvature vector H of M is continuous on M .*

Proof. Let $\{e_1, \dots, e_n\}$ be a local orthonormal frame of the tangent bundle. By definition,

$$H(x) = \sum_{i=1}^n B_x(e_i(x), e_i(x)).$$

Since each map $x \mapsto B_x(e_i(x), e_i(x))$ is continuous by the lemma, the finite sum above is continuous. Hence H is continuous on M . □

Definition 2.12 (Minimal surface). Let $M \subset \mathbb{R}^{n+k}$ be a n dimensional C^2 submanifold. We say that M is a *minimal surface* if for every $x \in M$ there exists a compact set $K_x \subset M$ with $x \in K_x$ such that for every compact perturbation $\{\phi_t\}$ of M (as defined in Definition 2.1) satisfying

$$\phi_t(y) = y \quad \text{for all } y \in M \setminus K_x,$$

we have

$$\mathcal{H}^n(\phi_t(K_x)) \geq \mathcal{H}^n(K_x)$$

for all sufficiently small t . In other words, M locally minimizes n -dimensional area.

Theorem 2.13. *Let M be a C^2 n -dimensional submanifold of $\mathbb{R}^{n+k}$. Then M is a minimal surface if and only if its mean curvature vector satisfies $\vec{H} \equiv 0$.*

Proof. We first prove that if M satisfies the local minimizing property, then $\vec{H} = 0$.

Fix $x \in M$. By definition, there exists a compact set $K_x \subset M$ with $x \in K_x$ such that for every perturbation $\{\phi_t\}$ of M with $\phi_t(y) = y$ for all $y \in M \setminus K_x$, we have

$$\mathcal{H}^n(\phi_t(K_x)) \geq \mathcal{H}^n(K_x)$$

for all sufficiently small t . Hence $t = 0$ is a minimum of the function

$$t \mapsto \mathcal{H}^n(\phi_t(K_x)),$$

and therefore

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\phi_t(K_x)) = 0.$$

Let X be any C^1 vector field on M supported in K_x , and consider the associated flow

$$\phi_t(y) = y + tX(y).$$

By the first variation formula,

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\phi_t(K_x)) = - \int_{K_x} \langle X, \vec{H} \rangle d\mathcal{H}^n.$$

Since the derivative vanishes for all such X , we obtain

$$\int_{K_x} \langle X, \vec{H} \rangle d\mathcal{H}^n = 0 \quad \text{for all vector fields } X \text{ supported in } K_x.$$

To conclude that $\vec{H} = 0$, let

$$K_{x,n} = \{y \in M : \text{dist}(y, M \setminus K_x) \geq 1/n\}.$$

Choose a smooth bump function f such that

$$f \equiv 1 \text{ on } K_{x,n}, \quad f \equiv 0 \text{ outside } K_{x,2n}.$$

Consider the vector field

$$X = f \vec{H}.$$

Substituting into the above identity gives

$$0 = \int_{K_x} \langle f \vec{H}, \vec{H} \rangle d\mathcal{H}^n = \int_{K_x} f |\vec{H}|^2 d\mathcal{H}^n.$$

Since $f \geq 0$ and $f = 1$ on $K_{x,n}$ and $\vec{H}$ is continuous, it follows that

$$\vec{H} = 0 \quad \text{on } K_{x,n}.$$

In particular, $\vec{H}(x) = 0$. Since x was arbitrary, we conclude that $\vec{H} \equiv 0$ on M .

Conversely, suppose that $\vec{H} \equiv 0$ on M . Let $\{\phi_t\}$ be a perturbation of M with compact support K , that is,

$$\phi_t(x) = x \quad \text{for all } x \in M \setminus K.$$

Let X denote the associated velocity vector field

$$X(x) = \left. \frac{\partial \phi_t}{\partial t}(x) \right|_{t=0}.$$

By the first variation formula we have

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\phi_t(K)) = - \int_K \langle X, \vec{H} \rangle d\mathcal{H}^n.$$

Since $\vec{H} \equiv 0$, it follows that

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\phi_t(K)) = 0.$$

Thus the area functional

$$t \mapsto \mathcal{H}^n(\phi_t(K))$$

has a critical point at $t = 0$.

However, the definition of a minimal surface requires the stronger conclusion that $t = 0$ is a local minimum. In general, this need not hold for perturbations supported in arbitrary compact sets. Nevertheless, for each $x \in M$ one can find a sufficiently small compact neighborhood K_x for which this property does hold.

To establish this requires analyzing the second variation

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \mathcal{H}^n(\phi_t(K_x)).$$

Although we do not derive the second variation formula here, its proof follows a similar strategy to that of the first variation formula. Using this formula together with a Poincaré inequality on sufficiently small balls, one can show that there exists a compact set $K_x \subset M$ containing x such that

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \mathcal{H}^n(\phi_t(K_x)) > 0$$

for all perturbations supported in K_x . Consequently, $t = 0$ is a local minimum of the area functional, and therefore

$$\mathcal{H}^n(\phi_t(K_x)) \geq \mathcal{H}^n(K_x)$$

for all sufficiently small t . This shows that M satisfies the local minimizing property, completing the proof. $\square$

We conclude this section by giving two examples of minimal surfaces.

Example 2.14. Let M be an n -dimensional hyperplane in $\mathbb{R}^{n+k}$, that is $M = x_0 + \text{span}\{v_1, \dots, v_n\}$, where $v_1, \dots, v_n$ are linearly independent vectors, $x_0 \in \mathbb{R}^n$. We claim that M is a minimal surface.

Extend $\{v_1, \dots, v_n\}$ to a basis $\{v_1, \dots, v_{n+k}\}$ of $\mathbb{R}^{n+k}$. Then the vectors $v_{n+1}, \dots, v_{n+k}$ form a basis of the normal space of M . Define normal vector fields on M by

$$\eta_i(x) = v_{n+i}, \quad i = 1, \dots, k.$$

These normal vector fields are constant. Consequently, their derivatives vanish, and by the definition of the second fundamental form we obtain $B \equiv 0$. Therefore the mean curvature vector also vanishes, $\vec{H} = 0$, and hence M is a minimal surface.

Example 2.15. The *catenoid* is the surface of revolution obtained by rotating the catenary $x = \cosh z$ about the z -axis. More precisely, we define $M \subset \mathbb{R}^3$ by the parametrization

$$\varphi(r, \theta) = (\cosh r \cos \theta, \cosh r \sin \theta, r), \quad r \in \mathbb{R}, \theta \in [0, 2\pi).$$

We cover M by two charts: $\varphi_1: \mathbb{R} \times (0, 2\pi) \rightarrow M$ and $\varphi_2: \mathbb{R} \times (-\pi, \pi) \rightarrow M$, both given by the same expression above.

We claim that M is a minimal surface, and we verify this by showing directly that $\vec{H} = 0$.

Step 1: Tangent vectors. Computing the partial derivatives of φ , we obtain

$$\partial_r \varphi = (\sinh r \cos \theta, \sinh r \sin \theta, 1), \quad \partial_\theta \varphi = (-\cosh r \sin \theta, \cosh r \cos \theta, 0).$$

One checks immediately that

$$\langle \partial_r \varphi, \partial_\theta \varphi \rangle = -\cosh r \sinh r \cos \theta \sin \theta + \cosh r \sinh r \sin \theta \cos \theta = 0,$$

so the two vectors are orthogonal. Their squared norms are

$$|\partial_r \varphi|^2 = \sinh^2 r + 1 = \cosh^2 r, \quad |\partial_\theta \varphi|^2 = \cosh^2 r.$$

Hence both vectors have norm $\cosh r$, and we define an orthonormal basis of $T_{\varphi(r, \theta)} M$ by

$$\tau_1 = \frac{\partial_r \varphi}{\cosh r} = \left(\frac{\sinh r \cos \theta}{\cosh r}, \frac{\sinh r \sin \theta}{\cosh r}, \frac{1}{\cosh r} \right), \quad \tau_2 = \frac{\partial_\theta \varphi}{\cosh r} = (-\sin \theta, \cos \theta, 0).$$

Step 2: Unit normal. We compute the unit normal N via the cross product:

$$\partial_r \varphi \times \partial_\theta \varphi = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \sinh r \cos \theta & \sinh r \sin \theta & 1 \\ -\cosh r \sin \theta & \cosh r \cos \theta & 0 \end{vmatrix} = (-\cosh r \cos \theta, -\cosh r \sin \theta, \cosh r \sinh r).$$

This vector has squared norm

$$\cosh^2 r \cos^2 \theta + \cosh^2 r \sin^2 \theta + \cosh^2 r \sinh^2 r = \cosh^2 r + \cosh^2 r \sinh^2 r = \cosh^2 r (1 + \sinh^2 r) = \cosh^4 r,$$

so its norm is $\cosh^2 r$. Dividing, we obtain the unit normal

$$N(\varphi(r, \theta)) = \frac{1}{\cosh^2 r} \left(-\cosh r \cos \theta, -\cosh r \sin \theta, \cosh r \sinh r \right) = \left(-\frac{\cos \theta}{\cosh r}, -\frac{\sin \theta}{\cosh r}, \frac{\sinh r}{\cosh r} \right).$$

Step 3: Computing $D_{\tau_1} N$. To differentiate N in the direction τ_1 , we choose the curve

$$\alpha(t) = \varphi(r + t, \theta),$$

which satisfies $\alpha(0) = \varphi(r, \theta)$, $\dot{\alpha}(0) = \partial_r \varphi = \cosh r \cdot \tau_1$. Then

$$N(\alpha(t)) = \left(-\frac{\cos \theta}{\cosh(r + t)}, -\frac{\sin \theta}{\cosh(r + t)}, \frac{\sinh(r + t)}{\cosh(r + t)} \right),$$

and differentiating with respect to t at $t = 0$ gives

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} N(\alpha(t)) &= \left(\frac{\cos \theta \sinh(r + t)}{\cosh^2(r + t)}, \frac{\sin \theta \sinh(r + t)}{\cosh^2(r + t)}, \frac{1}{\cosh^2(r + t)} \right) \Big|_{t=0} \\ &= \left(\frac{\cos \theta \sinh r}{\cosh^2 r}, \frac{\sin \theta \sinh r}{\cosh^2 r}, \frac{1}{\cosh^2 r} \right). \end{aligned}$$

Since $\dot{\alpha}(0) = \cosh r \cdot \tau_1$, the directional derivative in the direction τ_1 is

$$D_{\tau_1} N = \frac{1}{\cosh r} \cdot \left. \frac{d}{dt} \right|_{t=0} N(\alpha(t)) = \frac{1}{\cosh^3 r} (\cos \theta \sinh r, \sin \theta \sinh r, 1) = \frac{\tau_1}{\cosh^2 r}.$$

Hence

$$\langle D_{\tau_1} N, \tau_1 \rangle = \frac{|\tau_1|^2}{\cosh^2 r} = \frac{1}{\cosh^2 r}.$$

Step 4: Computing $D_{\tau_2} N$. We choose the curve

$$\beta(t) = \varphi(r, \theta + t),$$

which satisfies $\dot{\beta}(0) = \partial_\theta \varphi = \cosh r \cdot \tau_2$. Then

$$N(\beta(t)) = \left(-\frac{\cos(\theta + t)}{\cosh r}, -\frac{\sin(\theta + t)}{\cosh r}, \frac{\sinh r}{\cosh r} \right),$$

and differentiating at $t = 0$,

$$\left. \frac{d}{dt} \right|_{t=0} N(\beta(t)) = \left(\frac{\sin \theta}{\cosh r}, -\frac{\cos \theta}{\cosh r}, 0 \right).$$

Dividing by $|\dot{\beta}(0)| = \cosh r$,

$$D_{\tau_2}N = \frac{1}{\cosh r} \cdot \frac{d}{dt} \Big|_{t=0} N(\beta(t)) = \left(\frac{\sin \theta}{\cosh^2 r}, -\frac{\cos \theta}{\cosh^2 r}, 0 \right) = -\frac{\tau_2}{\cosh^2 r}.$$

Hence

$$\langle D_{\tau_2}N, \tau_2 \rangle = -\frac{|\tau_2|^2}{\cosh^2 r} = -\frac{1}{\cosh^2 r}.$$

Step 5: Conclusion. By definition, the mean curvature is

$$H = \langle D_{\tau_1}N, \tau_1 \rangle + \langle D_{\tau_2}N, \tau_2 \rangle = \frac{1}{\cosh^2 r} - \frac{1}{\cosh^2 r} = 0.$$

Therefore $\vec{H} \equiv 0$ on M , and hence the catenoid is a minimal surface.

References

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